# 10 Python Practice Problems with Solutions

Problems 1-7 are easy, 8-10 are medium. Try solving each problem yourself before reading the solution. Every solution below was run and the output shown is the real output.

### *Contents*

1. Reverse a String (Easy)
2. Check Palindrome (Easy)
3. Factorial of a Number (Easy)
4. Fibonacci Series (Easy)
5. Check Prime Number (Easy)
6. Count Vowels in a String (Easy)
7. Find the Largest and Smallest in a List (Easy)
8. Word Frequency Counter (Medium)
9. Remove Duplicates from a List (Medium)
10. Two Sum (Medium)

## Problem 1: Reverse a String

**Difficulty:** Easy

### *Problem*

Write a program that reverses a given string.
Example: 'python' -> 'nohtyp'

### *Solution*

```
def reverse_string(s):
    return s[::-1]

print(reverse_string("python"))
```

### *Output*

```
nohtyp
```

## Problem 2: Check Palindrome

**Difficulty:** Easy

### *Problem*

Write a function that checks whether a word is a palindrome (reads the same forwards and backwards), ignoring case.
Example: 'Madam' -> True

### *Solution*

```
def is_palindrome(word):
    word = word.lower()
    return word == word[::-1]

print(is_palindrome("Madam"))
print(is_palindrome("hello"))
```

### *Output*

```
True
False
```

## Problem 3: Factorial of a Number

**Difficulty:** Easy

### *Problem*

Write a function to find the factorial of a number n.
Example: 5 -> 120

### *Solution*

```
def factorial(n):
    result = 1
    for i in range(2, n + 1):
        result *= i
    return result

print(factorial(5))
```

### *Output*

```
120
```

## Problem 4: Fibonacci Series

**Difficulty:** Easy

### *Problem*

Print the first n numbers of the Fibonacci series.
Example: n = 8 -> 0 1 1 2 3 5 8 13

### *Solution*

```
def fibonacci(n):
    a, b = 0, 1
    series = []
    for _ in range(n):
        series.append(a)
        a, b = b, a + b
    return series

print(*fibonacci(8))
```

### *Output*

```
0 1 1 2 3 5 8 13
```

## Problem 5: Check Prime Number

**Difficulty:** Easy

### *Problem*

Write a function that returns True if a number is prime.
Example: 17 -> True, 18 -> False

### *Solution*

```
def is_prime(n):
    if n < 2:
        return False
    for i in range(2, int(n ** 0.5) + 1):
        if n % i == 0:
            return False
    return True

print(is_prime(17))
print(is_prime(18))
```

### *Output*

```
True
False
```

## Problem 6: Count Vowels in a String

**Difficulty:** Easy

### *Problem*

Count how many vowels are in a given string.
Example: 'Hello World' -> 3

### *Solution*

```
def count_vowels(text):
    return sum(1 for ch in text.lower() if ch in "aeiou")

print(count_vowels("Hello World"))
```

### *Output*

```
3
```

## Problem 7: Find the Largest and Smallest in a List

**Difficulty:** Easy

### *Problem*

Given a list of numbers, find the largest and smallest without using sorting.
Example: [4, 9, 1, 7] -> max 9, min 1

### *Solution*

```
def find_max_min(nums):
    largest = smallest = nums[0]
    for n in nums:
        if n > largest:
            largest = n
        if n < smallest:
            smallest = n
    return largest, smallest

print(find_max_min([4, 9, 1, 7]))
```

### *Output*

```
(9, 1)
```

## Problem 8: Word Frequency Counter

**Difficulty:** Medium

### *Problem*

Count how many times each word appears in a sentence (ignore case).
Example: 'the cat and the hat'

### *Solution*

```
def word_frequency(sentence):
    freq = {}
    for word in sentence.lower().split():
        freq[word] = freq.get(word, 0) + 1
    return freq

print(word_frequency("the cat and the hat"))
```

### *Output*

```
{'the': 2, 'cat': 1, 'and': 1, 'hat': 1}
```

## Problem 9: Remove Duplicates from a List

**Difficulty:** Medium

### *Problem*

Remove duplicate items from a list while keeping the original order.
Example: [1, 2, 2, 3, 1, 4] -> [1, 2, 3, 4]

### *Solution*

```
def remove_duplicates(items):
    seen = set()
    result = []
    for item in items:
        if item not in seen:
            seen.add(item)
            result.append(item)
    return result

print(remove_duplicates([1, 2, 2, 3, 1, 4]))
```

### *Output*

```
[1, 2, 3, 4]
```

## Problem 10: Two Sum

**Difficulty:** Medium

### *Problem*

Given a list of numbers and a target, return the indices of the two numbers that add up to the target.
Example: nums = [2, 7, 11, 15], target = 9 -> [0, 1]

### *Solution*

```
def two_sum(nums, target):
    seen = {}   # value -> index
    for i, n in enumerate(nums):
        need = target - n
        if need in seen:
            return [seen[need], i]
        seen[n] = i
    return None

print(two_sum([2, 7, 11, 15], 9))
```

### *Output*

```
[0, 1]
```